Day 1

Short Questions

1. A Vector Space V over the field of Real numbers $\mathbb{R}$ is a set of objects on which two primary operations are defined:

i) vector addition

ii) vector multiplication.

To qualify as vector space, the two closure properties required are:

i) Closure under addition: $u+v \in V$

ii) Closure under multiplication: $cu \in V$

2. A vector x in ~~$\mathbb{R}^n$~~ an n-dimensional real coordinate space, denoted as $\mathbb{R}^n$, is represented a a column matrix:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$$

~~It~~ It is a ordered n-tuple of real ~~nub~~ numbers written as,

$$x = (x_1, x_2 \ldots, x_n)^T.$$

In machine learning, each element x_i, represents a feature. For example, predicting house prices might require a vector in $\mathbb{R}^3$ where x_1 = square footage, x_2 = number of bedrooms, and x_3 = age of the property. Thus, a dataset with n features in mathematically represented as vectors in $\mathbb{R}^n$.

3. A subspace H is a subset of a vector space V that is itself a valid vector space. To prove that a subset is a subspace, the conditions are

i) $0 \in H$ (The zero vecor must exist within the subset)

ii) If $u, v \in H$, then $(u+v) \in H$ (closure under addition)

iii) If $u \in H$ and $c \in \mathbb{R}$, the $cu \in H$ (closure under scalar multiplication)

4. Linear Combinations of Vector. If a set of ~~vecto~~ vectors $\{v_1, \ldots, v_p\}$ in V, and a set of scalar weights $\{c_1, \ldots c_p\}$, a linear combination is the sum:

$$y = c_1 v_1 + c_2 v_2 + \ldots + c_p v_p = \sum_{i=1}^{p} c_i v_i$$

5. The span of a set of vectors, denoted Span $\{v_1, \ldots v_p\}$, is the continious collection of all possible linear combinations of those vectors.

6. Linear Independence: A set of vectors ~~is~~ are linearly independent if the only solution to the vector equation:

$$c_1 v_1 + c_2 v_2 + \ldots + c_p v_p = 0$$

7. A set of vectors are linearly dependant if The equation $(c_1 v_1 + c_2 v_2 + \ldots + c_p v_p)$ can be solved with non-zero weightts.

In machine learning, if a dataset has linearly dependent features (e.g., measuring

distance in both meters and kilometers). The model learns redundant information, leading to multi-colinearity and unstable model weight.

8. The Basis of a Vector Space: A basis for a ~~vis~~ V is set of vectors $B = \{b_1, \ldots, b_n\}$ that acts as a minimal, perfectly efficient coordinate system for that space. To form a basis, the two properties must a set of vectors satisfy:

i) It spans the space: $\text{Span}(B) = V$

ii) It is linearly Independent.

9. The ~~dimenti~~ dimension of a vector space is the number of vectors in any basis of the space:

$$\dim(V) = \text{number of basis vector}$$

10. The rank of a matrix equals the number of Linearly independent row vectors.

$$\text{rank}(A) = \dim(\text{Span of row Vectors})$$